

Optimizing Learning in University: Three Evidence-Based Study Strategies

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Introduction

As a university student, you spend many hours studying. Yet, many students are surprised and disappointed by their results. A common reason is that they rely on strategies that feel effective but are not: rereading a textbook chapter, for example, creates a sense of familiarity that is easily mistaken for real knowledge. Cognitive research has identified study techniques that produce more durable learning, yet students tend to underuse them. Self-report studies show that students favour rereading and highlighting over the methods that research supports (Miyatsu et al., 2018). Some reviews have responded by suggesting how students might improve the strategies they already prefer (Miyatsu et al., 2018) a more direct (one might even say, reasonable) route, however, is to adopt the strategies with the strongest evidence. This essay therefore asks how university students can optimize their learning, and answers by discussing three evidence-based strategies: pretesting, retrieval practice, and spaced practice. Effective studying is not simply a matter of effort, but of method.

Pretesting

The first strategy is pretesting: answering questions about the study material before any studying happens, even if the answers to the questions must be guessed. This might seem counterintuitive, but research indicates that this strategy improves later recall (Putnam et al., 2016). This activates any prior knowledge about the topic and prepares the student to connect new information with what is already known. Even incorrect guesses are useful as they make the student notice and remember the correct information once it appears.

Prequizzing is becoming easier to practice. (Putnam et al., 2016) suggests that before a study session (like a reading session or a lecture), a student can come up with questions about the topic and attempt to answer them, then return to those questions afterwards to check and correct the answers. However, many textbooks also include postreading quizzes. They can also be easily repurposed for pretesting. Thirdly, the rise of AI gives us new opportunities. A chatbot can create a quiz for you from your lecture slides or textbook chapter. It is even possible to prompt it to follow your precise specified needs such as difficulty level or quiz size. The main obstacle to this strategy is psychological rather than practical: it feels wrong to guess at material without any prior studying. Intuitively, students prefer to read first. Recognising that pretesting takes little time and prepares the memory helps to overcome this hesitation.

Retrieval Practice

The second strategy is retrieval practice: the reproduction or summarization of the study material after consuming it without looking at it. An implementation of this strategy is the RRR-method as described in (Putnam et al., 2016), Read, Recite, Review, where the student is supposed to first read the study material (or visit a lecture), secondly, write down a summary from memory and lastly, review their summary, that is compare it to the study material, see what they missed or misremembered. The retrieval would be the second step of this method. The important part here is that the student does not merely copy parts of the material but reproduces it from memory. This retrieval process has the advantage of both reactivating what they have learned as well as showing them what they have missed or forgotten.

From the common study strategy summarization this differs in a key way. The student will not use the material for summarization but merely write down what they remember. Thus the student will see what they did not fully understand or remember from the material, either during the recite step or when they later review, that is, compare their summary with the actual material. From the common study strategy of note-taking it also differs in the way that it doesn't happen while studying the material but afterwards and from memory. In its classical implementation, none of the common study strategies discussed in (Miyatsu et al., 2018) make use of retrieval. When highlighting, note-taking, summarization and outlining, a student merely alters or reproduces the material while studying.

As discussed in (Miyatsu et al., 2018), at least summarization and outlining can be altered in a way that makes use of retrieval by practicing these after studying the material from memory. For outlining, there is the caveat, however, that this is a skill that must be learned and practiced to be mastered.

Spaced Practice

The third strategy is spaced practice: this has nothing to do with how to study but with when to study. If a student uses spaced practice, they study for short periods of time, mix studying of different subjects and take considerable breaks between study sessions. In practice, this means studying each subject a little on most days and revisiting earlier topics in later sessions. The opposite of spaced practice is massed, crammed studying. According to (Putnam et al., 2016), for the same time spent studying, spaced practice gives much better results.

The typical example of cramming is a long study session the day or even the night before an exam. The difference in retention becomes even greater in the long term. So while

this might even give acceptable results on the day of the exam, for long-term learning, spaced practice is the much better study strategy.

Assuming university students are not merely studying to pass exams but are eager to learn and remember, spaced practice is the more effective strategy.

Conclusion

This paper examined three study strategies from (Putnam et al., 2016): pretesting, retrieval practice and spaced practice. The former two focus on retrieving prior and just acquired information; the latter is a strategy for allocating study time most efficiently.

Unlike the study strategies in (Miyatsu et al., 2018), the study strategies discussed in this paper are not chosen by popularity among students but by effectiveness. They do, however, have the disadvantage of needing more time in preplanning, a more conscious decision and commitment to stick to them as they may not be obvious or intuitive to students.

The technological advancements in AI can be of benefit to these study methods. For example, a student may feed their study material to a chatbot and have it create a pretesting quiz for them. In the RRR-method, they might ask an AI chatbot for feedback after the final review step of the method to make sure they didn't miss or misunderstand anything.

The strategies discussed in this paper do not mean that a student has to spend more time studying. With these more efficient strategies they should get better results with the same amount of time studying. Alternatively, if a student is already happy with their results, these study strategies should enable them to spend less time studying for a similar result. Any time saved can be spent as they wish.

References

Bibliography

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AI disclosure statement

An AI assistant was used extensively in the preparation of this essay, and this is disclosed in full. The assistant was opencode, running a DeepSeek language model. It drafted the initial text of the essay on the basis of the two assigned articles and the assignment brief, and it was also used to plan the structure of the essay and to discuss which study strategies to include. Every prompt sent to the assistant, and every reply it gave, are recorded in the project's public repository, together with a word-level attribution of the final text that shows which passages originated with the assistant and which were written or rewritten by the author. All sources cited in the essay were checked against the original articles, and the author reviewed and edited the final text and takes responsibility for its content.

Reflection

Writing this essay was more instructive than I expected. The two articles described study strategies that I have rarely used, and one in particular stood out: the read–recite–review method. Reading about it, I realised that I had planned to recite and review the articles after reading them, but never did, and I am now fairly sure that this is why I remember so little of their detail. The exercise also made me notice how often I choose study methods because they feel productive rather than because they work. In future, I want to try retrieval practice and spacing deliberately, and to pay attention to whether they feel harder at first. The writing itself was manageable, but I found it difficult to keep the text objective and to decide how much detail each strategy needed within the word limit.